SMART-ACSESS: SMART APPROACH IN DEVELOPING CAMPUS SECURITY AND SAFETY SYSTEMS

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ABSTRACT

This study presents SMART-ACSESS (Smart Approach in Developing Campus Security and Safety Systems), a unified information system and dashboard designed to enhance campus security operations at the University of the Philippines Los Baños Security and Safety Office (UPLB SSO). The system addresses key limitations of traditional security frameworks, including fragmented data sources, delayed incident reporting, and limited real-time decision support. By integrating multiple campus security systems into a centralized platform, SMART-ACSESS enables efficient data management, real-time monitoring, and improved operational responsiveness.

The proposed system utilizes modern technologies such as real-time Application Programming Interfaces (APIs), an extensible modular database architecture, and a generative AI-powered chatbot for statistical querying and reporting. These components support interactive dashboards, key performance indicator (KPI) tracking, and automated analytics for enhanced decision-making. Survey findings indicate that security personnel prioritize real-time monitoring, vehicle tracking, and identification systems, while chatbot functionalities are considered supplementary.

Usability evaluation using the System Usability Scale (SUS), grounded in the Technology Acceptance Model (TAM), yielded an average score of 73.33, interpreted as "Good," indicating above-average usability and general user acceptability.

Overall, SMART-ACSESS provides a scalable and adaptable framework for campus security management by integrating data visualization, analytics, and AI-driven support. The study highlights the value of centralized, data-driven systems in improving situational awareness, optimizing resource allocation, and strengthening campus safety initiatives.

I. INTRODUCTION

Effective campus security management is a difficult challenge, particularly in handling vast amounts of data, ensuring real-time information accessibility, and maintaining transparency across operations. Traditional security systems often fall short, particularly for implementing unified dashboards, data visualizations, and extensible database systems to meet the dynamic needs of modern campus environments. Issues such as delayed incident reporting, further exacerbate these challenges, preventing security teams from responding proactively to emerging threats. These challenges highlight the need for a comprehensive and adaptable security information system that optimizes data usability, accessibility, and visualization.

The proposed system addresses the operational needs of the University of the Philippines Los Baños Security and Safety Office (UPLB SSO) under the Office of the Vice Chancellor for Community Affairs (OVCCA). As the primary unit responsible for ensuring the safety, security, and welfare of the UPLB community, the UPLB SSO faces increasing demands for quick decision-making, effective resource management, and proactive incident and/or violation resolution. Given the growing complexity of security challenges, the office requires a technological solution capable of streamlining operations and improving response efficiency. The study proposes a Smart Approach to Campus Safety and Security System (SMART-ACSESS) through a campus security unified information system and dashboard that integrates existing security system databases with modernized technologies such as real-time Application Programming Interfaces (APIs), extensible database for multi-system integration, and generative AI-powered chatbots.

The system aims to enhance the office's capability to monitor, analyze, and respond to security issues in real time. It includes features such as real-time analytics for tracking key performance indicators (KPIs) and advanced data visualization tools to support decision-making. Furthermore, its modular database design ensures compatibility with future technological advancements, offering a future-proof solution tailored to the system requirements of UPLB SSO.

By implementing this unified system, the UPLB SSO will be better equipped to fulfill its mandate of maintaining campus safety and security. The project emphasizes accessibility, adaptability, and efficiency, aligning with UPLB's broader vision of leveraging technology for institutional excellence. This study not only addresses immediate operational gaps but also provides a sustainable framework for evolving security needs, ensuring that UPLB remains a safe and secure environment for its community.

Significance of the Study

The proposed system's significance lies in its potential to transform campus security management by:

1. Improved decision-making through enhanced data visualization and KPI analytics through dashboarding.
2. Enhancing efficiency though real-time APIs and system database integrations to streamline and centralize data management processes.
3. Extensible features by implementing modular features that enables the system to adapt and integrate with future technological and operational needs.

General Objectives

To design a campus security unified information system and dashboard that facilitates features including data visualization through dashboarding, real-time Application Programming Interface (APIs) for system databases integration, and generative AI chatbot for enhanced and efficient operations.

Specific Objectives

Specifically, the study aims to achieve the following objectives:

- Dashboard and data visualization
- Extensible and integrated database
- Generative AI-powered chatbot

Scope and Limitations

This study focuses on designing a campus security unified information system and dashboard specifically for campus environments.

- The study covers the design and development of a campus security unified information system and dashboard tailored for campus environments.
- The system operates under the assumption of users' basic technical proficiency and availability of campus infrastructure supporting real-time database integration.
- It utilizes accurate and consistent data from integrated campus security database systems for KPI tracking and analysis.
- The system features an extensible and modular architecture for integrating additional databases within the defined framework, excluding external or cross-institutional systems.
- Functionality includes online operations, with a generative AI chatbot limited to narrative and statistical outputs, excluding cloud-based deployment, offline functionality, and other generative media outputs.

Organization of the Study

The organization of the study focuses on the University of the Philippines Los Baños Security and Safety Office (UPLB SSO) under the Office of the Vice Chancellor for Community Affairs (OVCCA). The study is structured to align with the operational framework, existing infrastructure, and evolving needs.

UPLB SSO oversees campus security operations, including surveillance, incident reporting, and resource management. With increasing data complexity and the need for quick decision-making, the office requires a unified, technologically aligned system to streamline operations and improve response times. The study's proposed a unified information system and dashboard that integrates existing security system databases with modernized technologies such as real-time Application Programming Interfaces (APIs), extensible database for multi-system integration, and generative AI-powered chatbots. These will enhance the office's capability to monitor, analyze, and respond to security issues effectively.

II. Review of Related Literature and Research

A. Dashboarding and Data Visualization

Dashboarding has emerged as an essential tool for decision-making in organizations, providing a centralized platform for visualizing complex data. By translating raw data into intuitive graphs, charts, and metrics, dashboards enhance comprehension and enable users to identify trends, anomalies, and key performance indicators (KPIs). The integration of robust database systems into dashboards allows for real-time updates, fostering data-driven strategies across sectors such as business intelligence, public safety, and resource management. Effective dashboards are characterized by user-centric design principles and dynamic data visualization capabilities that cater to diverse organizational needs.

Dashboarding, data analytics, and visualization are transformative in enhancing decision-making and optimizing security frameworks across various domains. These technologies allow for real-time monitoring, predictive analysis, and automated insights, helping authorities and organizations streamline operations based on data-driven strategies.

Big data analytics plays a central role in extracting meaningful patterns from complex datasets. Bachechi, Po, and Rollo [1] illustrated this in their study on the Trafair Traffic Dashboard (TTD), which integrates sensor data, anomaly detection, and GIS-based mapping to analyze urban traffic and air pollution. Their findings highlight how visualization tools enhance urban mobility by tracking congestion patterns and identifying environmental impacts. This methodology can be extended to security systems, where similar dashboards could support live surveillance, monitor security trends, and detect anomalies in access control.

AI-powered data visualization provides another layer of efficiency in policy and governance. Lino Ferreira da Silva Barros et al. [2] leveraged machine learning models to classify academic research according to Sustainable Development Goals (SDGs), integrating AI-powered classification into an interactive PowerBI dashboard. This approach underscores the importance of automated decision support systems, showing how AI models can categorize complex datasets and improve funding allocations. Within the realm of campus security, AI-driven dashboards could similarly aid in assessing risk levels, optimizing emergency response strategies, and automating security breach detection.

Predictive analysis, particularly through automated machine learning (AutoML), further strengthens data-driven systems. Rezaul et al. [3] compared manual machine learning techniques with AutoML frameworks, finding that automated models consistently performed with greater accuracy and efficiency. Their study demonstrated how machine learning could refine decision-making processes by reducing manual intervention while enhancing predictive capabilities. In campus security, AutoML could process historical access logs and security incidents to forecast potential threats, thereby improving the efficiency of surveillance and response mechanisms.

Meta-modelling also contributes to refining visualization structures, ensuring usability and adaptability in dashboard designs. Vázquez-Ingelmo et al. [4] examined meta-modelling approaches for structuring dashboard frameworks, showing how expert validation helps refine information representation, user interaction, and data visualization efficiency. Their findings suggest that an intuitive meta-model improves the coherence and accessibility of complex dashboards, ensuring that decision-makers can interpret data quickly and effectively. In security applications, well-structured dashboards could integrate real-time security alerts, threat level assessments, and intelligent response coordination to create a seamless safety infrastructure.

Integrating dashboarding, data analytics, and visualization enables security systems to evolve into proactive, AI-powered frameworks capable of real-time surveillance and predictive

risk assessment. These studies collectively reinforce that dynamic dashboards, automated analytics, and structured modelling drive efficiency in security management, decision-making, and emergency response systems. The adoption of these technologies in campus security frameworks, such as SMART-ACSESS, could significantly enhance safety measures by providing real-time threat monitoring, AI-driven predictive analytics, and intuitive security dashboards for optimized risk assessment and response execution.

B. API-Enabled Technologies

The rapid advancement of Application Programming Interface (API)-enabled technologies has transformed the way data is accessed, shared, and utilized in various industries. APIs serve as the backend of modern applications, enabling seamless integration of disparate systems and real-time data exchange. In the context of data analytics, APIs provide researchers and organizations with tools to retrieve, process, and analyze vast datasets efficiently, fostering a deeper understanding of complex systems. These technologies have found applications in fields like health informatics, IoT, and environmental monitoring, where dynamic and interoperable solutions are crucial for actionable insights.

Application programming interfaces (APIs) are fundamental to modern software architecture, streamlining data exchange and interoperability between systems. Studies on API design, security, evolution, and analytics provide valuable insights into optimizing these frameworks for enhanced functionality and protection. The integration of dashboarding, data analytics, and visualization improves API usability, security monitoring, and system evolution.

Bogner, Kotstein, and Pfaff [5] examined RESTful API design rules and their impact on understandability. Their controlled web-based experiment assessed comprehension based on rule violations across API snippets. For 11 of the 12 tested rules, violations significantly reduced readability, with CRUD naming conventions in URIs being most detrimental. Their findings suggest that structured API design, adherence to best practices, and standardized visualization tools improve API accessibility and usability.

Yin, Song, and Zong [6] advanced API security by introducing a two-stage code mining approach, improving the detection of API call sequence violations. Their methodology extracted frequent API sequences, refined security rule definitions, and visualized API relationships through graph-based aggregation techniques. By integrating data visualization and automated security monitoring, their approach enhances API governance frameworks, providing real-time security tracking, anomaly detection, and threat response mechanisms.

Lercher et al. [7] investigated microservice API evolution, analyzing strategies developers employ to manage updates while maintaining system stability. Their study highlighted API versioning, backward compatibility, and event-driven communication, stressing that visualization tools can assist in tracking API dependencies, assessing consumer impact, and documenting API evolution. The integration of dashboarding tools can mitigate risks associated with API changes, supporting efficient communication across development teams.

Khanyi et al. [8] conducted a systematic review of API applications within small and medium enterprises (SMEs). Their findings revealed that APIs and data networks improve operational efficiency, accelerating business processes by 12% while reducing operational costs by 8%. However, security vulnerabilities remain a challenge, with API-related cyberattacks increasing by 400% in Q1 2023. Their study underscores the importance of dashboarding and security visualization, allowing SMEs to monitor API activity, detect vulnerabilities, and enhance regulatory compliance through predictive analytics and security dashboards.

Wu et al. [9] explored the future of API analytics, emphasizing how machine learning and natural language processing (NLP) enhance API documentation, usage insights, and developer experience. Their study highlights the importance of visualization tools for API analytics, enabling developers to extract API usage patterns, optimize reusability, and detect inefficiencies in software development workflows.

The integration of data analytics, visualization techniques, and dashboarding tools significantly strengthens API governance, security monitoring, and usability optimization. These studies reinforce that structured API design, automated security visualization, and interactive dashboards improve API understandability, security tracking, and operational efficiency. By adopting data-driven insights and visualization methods, APIs can evolve into more robust, transparent, and scalable infrastructures while addressing security risks and compliance challenges.

C. Generative AI – Statistical Chatbot Querying

Generative AI-powered chatbots have revolutionized how users interact with complex datasets by offering natural language interfaces for querying and report generation. These advanced systems use AI to interpret user inputs, process underlying data, and produce statistical insights in real-time. In fields like customer support, business analytics, and academic research, these chatbots facilitate seamless access to information, reducing the need for specialized technical expertise. Their ability to generate coherent, contextually relevant reports positions them as valuable tools in streamlining decision-making and enhancing user engagement with data systems.

Generative AI and chatbot systems have significantly evolved, offering new possibilities for interactive policy communication, intelligent data visualization, and API-based chatbot orchestration. The reviewed studies highlight how retrieval-augmented generation (RAG), model-driven engineering (MDE), and AI-powered natural language interfaces contribute to more efficient chatbot systems.

The iTANONG project, developed by DOST-ASTI [10], leverages artificial intelligence and natural language processing (NLP) to revolutionize information accessibility by enabling users to query relational databases in everyday language. Through the integration of APIs and NLP models, the system facilitates seamless, real-time data retrieval without requiring technical expertise, promoting inclusivity and accessibility. While primarily focused on NLP-driven queries, iTANONG has the potential to enhance decision-making by integrating dynamic dashboards for real-time data visualization. Its multilingual capabilities, including Filipino and Taglish, further expand user reach and allow for the generation of localized statistical reports and actionable insights. This technology holds applications across various industries, such as finance, healthcare, and e-commerce, where intuitive data analysis is critical. Overall, iTANONG exemplifies the transformative potential of AI in democratizing access to

information, emphasizing inclusivity, linguistic diversity, and user-centric design to empower Filipinos and address real-world challenges. Its continued evolution will likely pave the way for broader AI-driven solutions catering to local needs.

Yun, Yun, and Xue [11] introduced an AI-powered citizen-government interaction system, integrating retrieval-augmented generation (RAG) with large language models (LLMs) to improve policy query responses. Their study processed 200 policy documents from Chinese and US governments, embedding text for optimized semantic search via FAISS. The system demonstrated 85.58% accuracy for Chinese policies and 90.67% accuracy for US policies, proving highly effective in handling numerical data queries. However, challenges included retrieving correct regional policies and extracting text from complex PDFs. The findings suggest that AI-powered chatbots could streamline security policy dissemination, enhancing campus safety protocols and emergency response frameworks.

Kavaz, Puig, and Rodríguez [12] conducted a scoping review on chatbot-based natural language interfaces (V-NLIs) for data visualization. Their study found that 70% of V-NLIs use basic tabular data, while 30% handle hierarchical or temporal data types. Most systems relied on fixed visual mappings (50%) or rule-based mappings (20%), while only 10% applied AI-driven visual mapping. Follow-up queries were limited, as only 20% supported multi-turn dialogues. The findings underscore the need for AI-enhanced security dashboards, allowing multi-turn conversational interactions for real-time incident analysis, access control monitoring, and predictive risk assessment.

Ma, You, and Wang [13] proposed BotSwagger, an OpenAPI extension for semi-automated chatbot generation using model-driven engineering (MDE). The BOTEN system converted BotSwagger specifications into functional chatbots, achieving 96% accuracy in selecting the correct chatbot based on user queries. Their approach significantly reduces manual effort in chatbot integration while improving API orchestration for complex services. This methodology could support campus security frameworks, enabling multi-chatbot coordination for emergency response automation and surveillance insights.

Harahap, Saad, and Ubaidullah [14] conducted a comprehensive bibliometric review of chatbot research, analyzing 955 publications from 2010–2024. Their study identified shifts from rule-based chatbots to AI-driven conversational systems, with thematic trends focused on security, education, and healthcare applications. The rise of generative AI models, such as GPT and BERT, has expanded chatbot functionality, improving context awareness and adaptive responses. Their findings suggest that security chatbots must integrate ethical safeguards, ensuring privacy-preserving AI models for sensitive user interactions.

The integration of AI-powered chatbots, retrieval-augmented data systems, and interactive security dashboards presents transformative opportunities for policy communication, security intelligence, and API-driven chatbot frameworks. The reviewed studies collectively reinforce that natural language interfaces, structured API chatbot generation, and AI-enhanced visualization are crucial for developing robust, adaptive chatbot systems capable of real-time threat monitoring, security compliance automation, and multi-agent coordination.

III. METHODOLOGY

The study frameworks provide a comprehensive foundation for understanding the theoretical and practical aspects of designing a campus security unified information system and dashboard.

Specifically, this study aims to design a **campus security unified information system and dashboard** that leverages advanced features such as:

- **Analytics and Data Insights**
  - Develop a dashboard with real-time data visualization to track Key Performance Indicators (KPIs) relevant to campus security operations.
- **Database and API Integration**
  - Implement an extensible database architecture to handle modular and dynamic data storage.
  - Implement APIs that enable database-driven system functionalities, ensuring data consistency and accessibility across systems.
- **Generative AI Chatbot**
  - Incorporate a generative AI-powered chatbot for automated statistical report generation and query resolution.
- **User and System Management**
  - Establish a secure user registration system with password-driven authentication and access controls to manage user roles and rights effectively.
- **Operational Capabilities**
  - Online operational functionality to maintain uninterrupted security operations.
  - Design modular features for easy integration of future database systems and upgrades.
- **Extensible System Database**
  - Information system database that supports future system database add-ons, integration and compatibility with evolving campus security systems.

Theoretical Framework

The theoretical framework draws on established principles such as Unified Systems Theory, which highlights the importance of integrating subsystems for operational efficiency, and the Technology Acceptance Model (TAM), which focuses on user acceptance as a critical success factor. These theories inform the development of a cohesive and user-friendly platform tailored to the security needs of the UPLB SSO.

The theoretical framework grounds the study in established theories and principles that explain the dynamics of unified information systems, API-enabled technologies, data visualization, and AI-driven solutions.

1. **Unified Systems Theory –** This theory emphasizes the integration and coordination of multiple database systems into a cohesive whole. In the context of campus security management, the theory underscores how unified information systems improve efficiency by centralizing data processing, communication, and resource management. This aligns with the study's goal of creating a seamless system for real-time data exchange and visualization.
2. **Technology Acceptance Model (TAM) –** TAM provides insights into user acceptance of technology, focusing on perceived usefulness and ease of use as critical factors. For the proposed system, TAM is applicable to ensuring that users (e.g., security

personnel and administrators) find the dashboard intuitive and the AI chatbot effective for their tasks.

3. **Artificial Intelligence and Decision-Making Theories** – Generative AI, grounded in machine learning and natural language processing theories, demonstrates how AI enhances decision-making by automating complex tasks like statistical report generation and predictive analytics. These theories support the incorporation of generative AI into chatbot features, ensuring accurate and relevant outputs.

Conceptual Framework

On the other hand, the study's conceptual framework serves as a blueprint for the system's design and implementation, detailing the relationships between inputs, processes, and outputs. This framework ensures that the system is not only effective in addressing current security challenges but also adaptable for future enhancements.

As shown in the Figure No. 1, the Conceptual Diagram represents the system's high-level logic and serves as the conceptual framework. It begins with key inputs such as technological resources (e.g., APIs, extensible databases, AI algorithms), security requirements (like audit trails and access controls), and user needs (particularly for data visualization and operational analytics). These inputs feed into the processes, which include system design and development, the integration of real-time data exchange mechanisms, the implementation of AI-powered features (such as a generative chatbot for querying and reporting), and user management for access and activity control. These processes lead to the outputs: a unified security dashboard for real-time insights, improved operational efficiency, and an extensible, modular system for future enhancements.

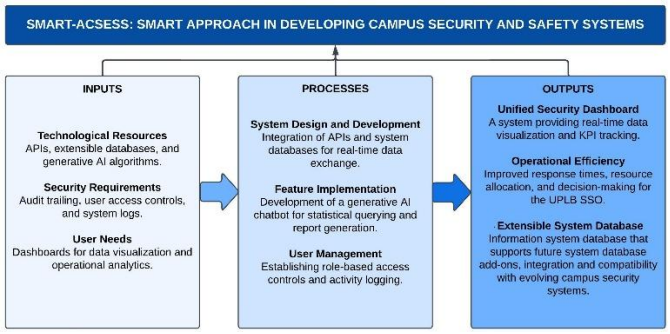


Figure No. 1: Conceptual Framework

System Architecture Diagram

Based on Figure No. 2, the System Architecture illustrates the technical structure of the proposed system, highlighting how multiple campus security systems are integrated into a unified platform. It features an API Modular Database layer that facilitates the connection and integration of various systems, including UPLB's AMIS and UVISS, as well as subsystems for vehicle monitoring, facial recognition for curfew violations, campus traffic management, parking violation monitoring, and other external databases. These integrated data sources are consolidated within a unified database, which serves as the central repository for all processed information. The centralized data is then accessed by core components such as the SMART-ACSESS dashboard for visualization, user management, and the generative AI module for statistical report queries, enabling real-time insights and data-driven decision-making.

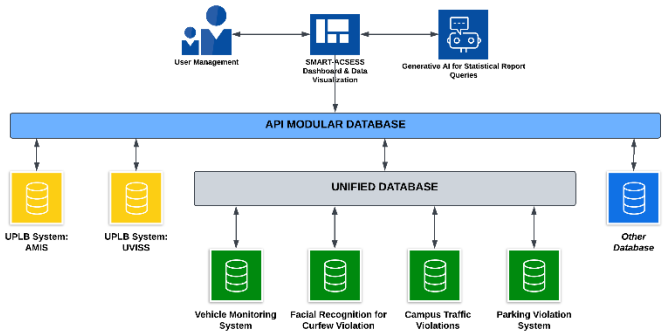


Figure No. 2: System Architecture

Use Case Diagram

The Use Case Diagram, as shown in Figure No. 3, illustrates the interactions between the Campus Security Unified Information System and its primary actors, namely the End-User (OVCCA/UPLB SSO) and the Admin. The End-User can perform key functions such as User Authentication, View Dashboard, Search and Filter Data, Generate AI-based Statistical Reports, and Export Reports, while the Admin manages user accounts and system configurations. Central to these operations is the "Retrieve Integrated Data" function, which enables the system to access and consolidate information from multiple external systems, including UVISS, AMIS, facial recognition for curfew violations, traffic violations, vehicle monitoring, parking violations, and other databases. These external systems are treated as independent sources that provide data to the unified system. Overall, the diagram highlights how the proposed system serves as a centralized platform that integrates diverse campus security systems to support real-time monitoring, analytics, and data-driven decision-making.

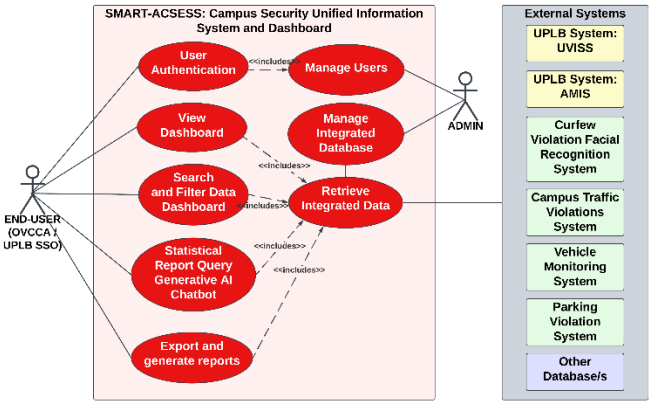


Figure No. 3: Use Case Diagram

Data Architecture Diagram

The Data Architecture diagram, as shown in Figure No. 4, illustrates the flow and processing of information across the proposed system. It begins with multiple campus security data sources accessed through API Gateways, which are consolidated and managed within the API Modular Database Integration layer. The integrated data is then stored in a centralized database (PostgreSQL), serving as the system's primary data repository. From this central storage, data undergoes processing and analysis through ETL mechanisms and analytics components, enabling dashboard visualization, KPI tracking, and AI-powered report generation. These outputs are then delivered through the unified system interface, providing users with a centralized and accessible platform for monitoring and decision-making. Overall, the architecture presents a structured and scalable data

pipeline that integrates data management, processing, and intelligent analytics into a cohesive campus security solution.

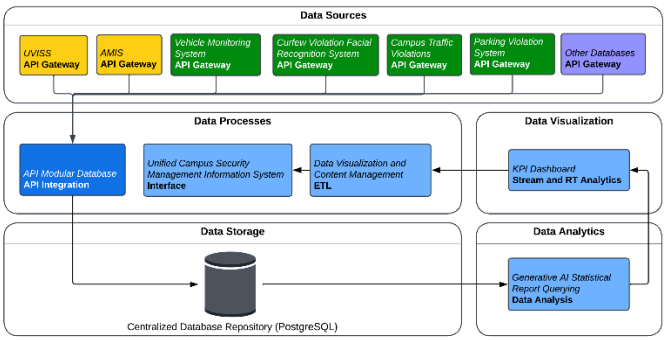


Figure No. 4: Data Architecture Diagram

Applications Architecture Diagram

The Applications Architecture Diagram as shown in the Figure No. 5, is divided into three main layers: the Presentation Layer, the Application Layer, and the Data Layer. The Presentation Layer includes the user interface, enabling interaction through web or desktop applications. The Application Layer handles core functionalities such as user management, user authentication, and access control management. It also includes modules for data visualization, including KPI dashboards, and a generative AI-powered statistical reporting tool. Additionally, this layer supports modular database management, allowing extensibility through API endpoints and connections with external systems. The Data Layer consists of a PostgreSQL database, which stores and manages data used across the system. Below these layers, the architecture integrates with multiple external systems and services, such as UPLB's UVISS and AMIS, as well as the curfew violation system (facial recognition), vehicle monitoring system, campus traffic violations, and parking violation system. These services provide input and retrieve output via API integration.

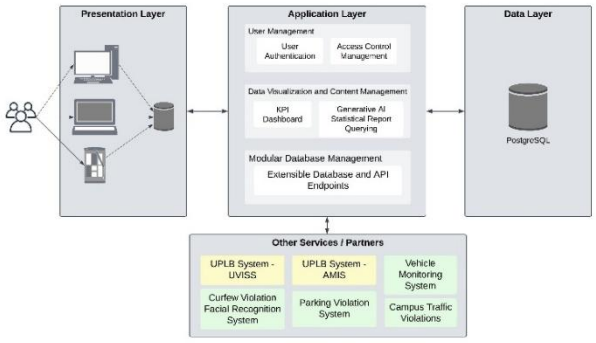


Figure No. 5: Application Architecture Diagram

IV. FINDINGS

i. SMART-ACSESS System User Interface

To further illustrate the developed SMART-ACSESS system, this section presents selected screenshots of the implemented interface and core functionalities. These visuals provide a clearer representation of the system's user interface, dashboard features, data visualization components, and AI-powered chatbot interactions. The screenshots aim to complement the system design and architecture by demonstrating how the proposed features are realized in the actual implementation, highlighting the system's usability and practical application.

Figure No. 6 presents the login interface of the SMART-ACSESS system. The interface allows users to securely access the platform using their authorized-UP Mail accounts, subject to administrative approval. It provides authentication options, including standard credential login and Google-based sign-in, ensuring secure and role-based access to system modules.

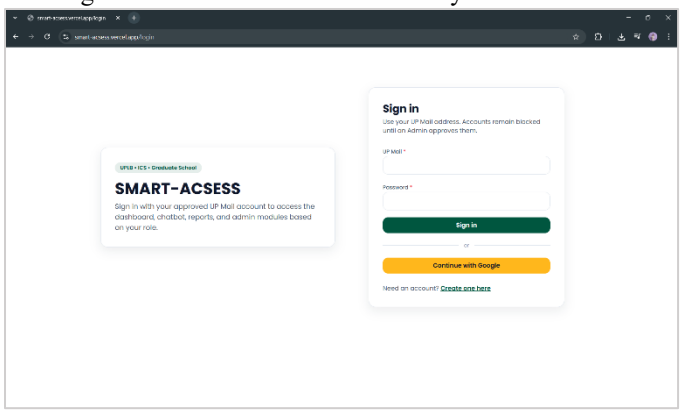


Figure No. 6: SMART-ACSESS Login Page

Figure No. 7 illustrates the dashboard reporting module of SMART-ACSESS. This interface enables users to generate reports by selecting specific databases, tables, and filters. It supports customizable data visualization and export functionality, allowing users to produce structured reports and graphical representations for decision-making purposes.

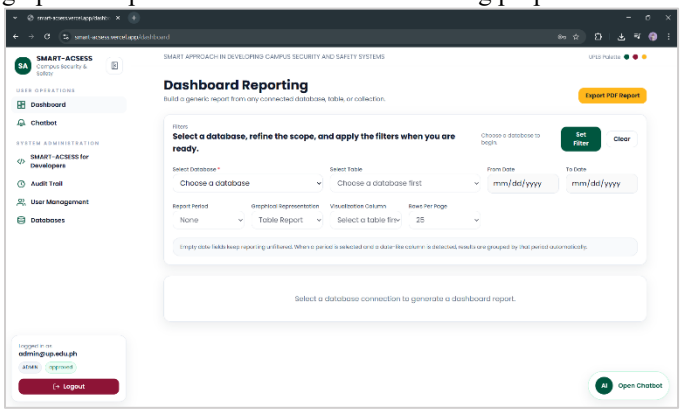


Figure No. 7: SMART-ACSESS Dashboard Module

Figure No. 8 shows the generative AI-powered Campus Safety Chatbot of SMART-ACSESS. The chatbot allows users to query campus security data using natural language, supporting both English and Tagalog inputs. It also features saved conversations and suggested queries, enhancing accessibility and ease of use for retrieving insights and reports.

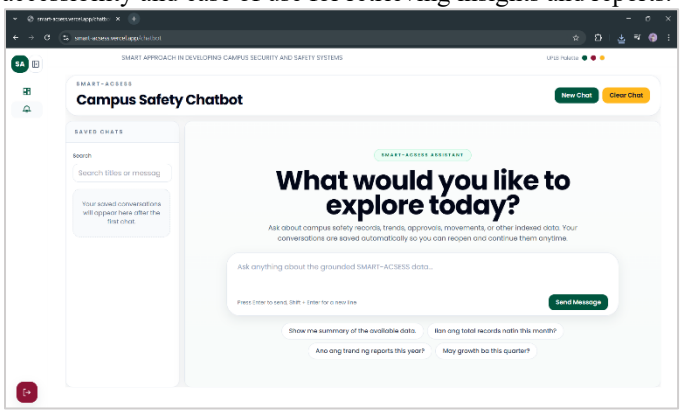


Figure No. 8: SMART-ACSESS Chatbot Module

Figure No. 9 and 10 presents a sample interaction with the SMART-ACSESS Campus Safety Chatbot, demonstrating its capability to process natural language queries and generate data-driven responses. In this example, the user inquiries about the number of sticker applications reviewed by specific personnel in Tagalog Language. The system retrieves relevant records from the integrated database and provides a summarized response, including total record count and supporting details. Additionally, the chatbot presents structured outputs such as tabular results and allows export functionality, highlighting its role in facilitating efficient data retrieval, reporting, and decision support.

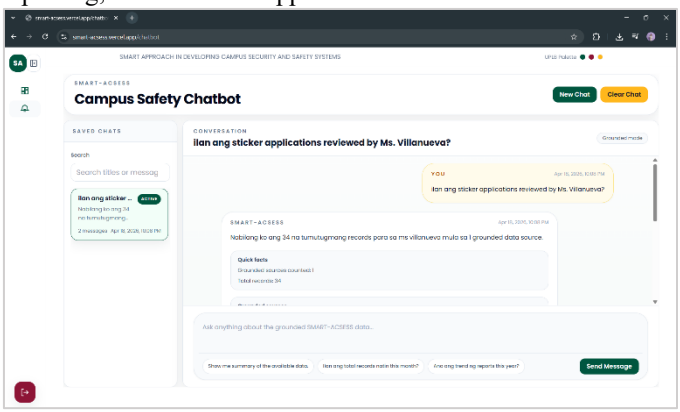


Figure No. 9: SMART-ACSESS Chatbot Sample Response

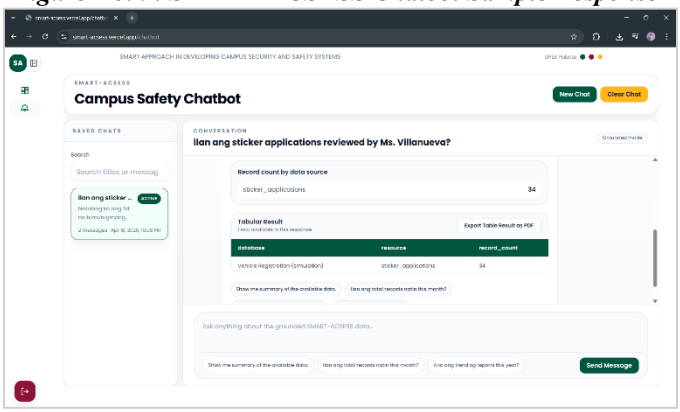


Figure No. 10: SMART-ACSESS Chatbot Sample Response (continuation)

Figure No. 11 presents the SMART-ACSESS for Developers module, which serves as an admin-restricted interface for accessing system integration documentation. It provides guidelines on API usage, authentication, and data handling, and includes a downloadable PDF for external developers integrating with the system.

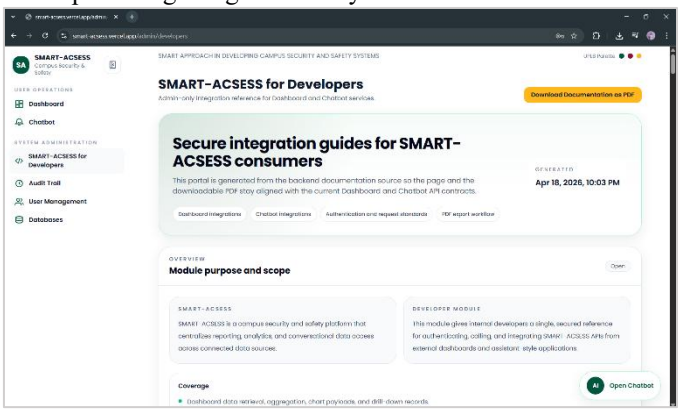


Figure No. 11: SMART-ACSESS Developers Guide

Figure No. 12 illustrates the audit trail module of SMART-ACSESS. This interface enables administrators to monitor and review system activities, including user actions, database interactions, and report generation. It supports filtering and tracking mechanisms to ensure transparency, accountability, and system security.

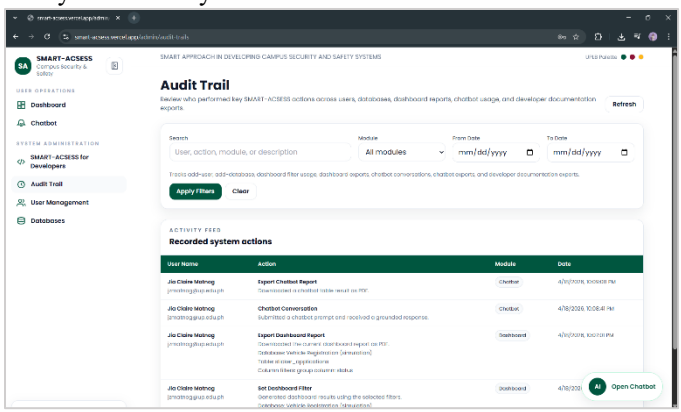


Figure No. 12: SMART-ACSESS Audit Trail Module

ii. SMART-ACSESS KPI Pre-Survey Results

The consolidated survey responses provide a comprehensive overview of campus security personnel's perspectives on the implementation of SMART-ACSESS: Smart Approach in Developing Campus Security and Safety Systems. The data highlights key areas of concern, essential functionalities, preferred reporting formats, and critical security performance indicators. By analyzing responses from security personnel across various experience levels, these findings shed light on current challenges, essential improvements, and recommendations for an effective centralized security dashboard. The survey results serve as a foundation for designing an optimized campus security framework that integrates real-time monitoring, automated tracking, and AI-driven analytics to enhance overall campus safety.

Security Personnel Profile and Experience

As shown in Table 1, the majority of survey respondents (17 out of 18) identified as security personnel, with no representation from campus administrators, IT staff, faculty, or students. This skewed distribution suggests that direct stakeholders in campus security management are actively engaged in shaping the development of the centralized security dashboard. Additionally, the data indicates that the majority of personnel have 6–10 years of experience (9 responses), implying that the dashboard should incorporate features aligned with experienced security professionals' operational needs while remaining intuitive for newer personnel (1–3 years, 2 responses).

NO.	POSITION/ROLE	RESPONSES
1	Security Personnel	17
2	System Administrator	0
3	Campus Administrator	0
4	IT Staff	0
5	Faculty Member	0
6	Student Representative	0
7	Other	1

Table 1. KPI Survey: General Information

Primary Goals of the Centralized Security Dashboard

Respondents highlighted in Table 2 that real-time monitoring and incident response (13 responses) and enhanced safety for students, staff, and visitors (15 responses) as the most critical objectives. Other top priorities include tracking curfew violators (12 responses) and reducing traffic/parking violations (11 responses). While generative AI chatbot integration received fewer endorsements (6 responses), this suggests that automated query handling could complement security operations rather than replace direct monitoring efforts

NO	How critical is the following functionality for your role?	RESPONSES					TOTAL
		Not Critical	Uncritical	Neutral	Critical	Extremely Critical	
1	Real-time vehicle monitoring	4	4	1	0	8	**17**
2	Traffic/parking violation reporting	3	4	1	3	6	**17**
3	Facial recognition for curfew violators	2	4	2	1	8	**17**
4	Generative AI chatbot for user queries and reports	8	1	1	1	4	**15**
5	Statistical reporting of security incidents	3	2	3	3	6	**17**

Table 2. KPI Survey: Objectives and Requirements

Critical Functionalities for Campus Security Operations

The evaluation of dashboard functionalities revealed in Table 3 that the real-time vehicle monitoring (8 responses rated it as extremely critical) as a top priority, reinforcing the need for an efficient vehicle tracking system within SMART-ACSESS. Similarly, facial recognition for curfew violators (8 responses rated it as extremely critical) indicates that biometric identification systems could streamline violation management. On the other hand, generative AI chatbot capabilities were rated as less critical (8 responses marked it as 'not critical'), implying a lower dependency on AI-assisted query resolution compared to traditional security reporting systems.

NO.	What are the primary goals of the centralized campus security dashboard?	RESPONSES
1	Real-time monitoring and incident response	13
2	Reduction of traffic/parking violations	11
3	Efficient tracking of curfew violators	12
4	Enhanced safety for students, staff, and visitors	15
5	Integration of generative AI chatbot for user inquiries	6
6	Other	3

Table 3. KPI Survey: Critical Functionalities

Key Performance Indicators (KPIs) for Integrated Databases

Security personnel identified critical metrics across vehicle registration, monitoring, traffic violations, and facial recognition systems. The highest-priority KPIs included number of vehicles registered (12 responses) on Figure 13, frequency of system updates for student data (11 responses) on Figure 14, number of vehicles tracked daily (12 responses) on Figure 15, and accuracy rate of the facial recognition system for curfew violations (12 responses) on Figure 18. Additionally, security staff emphasized the importance of monitoring traffic violations recorded per day/week/month (13 responses) on Figure 16 and parking violation detection (12 responses) on figure 17 to improve campus safety enforcement.

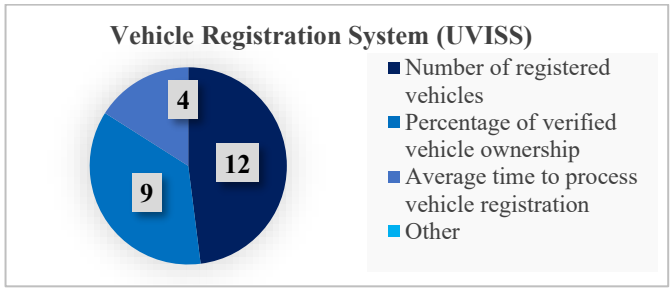


Figure 13. KPI Result for Vehicle Registration System UVISS

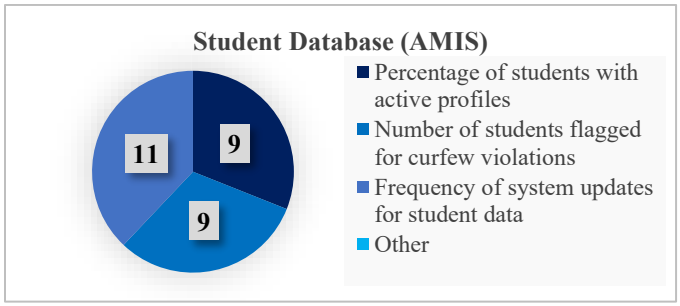


Figure 14. KPI Result for Student Database AMIS

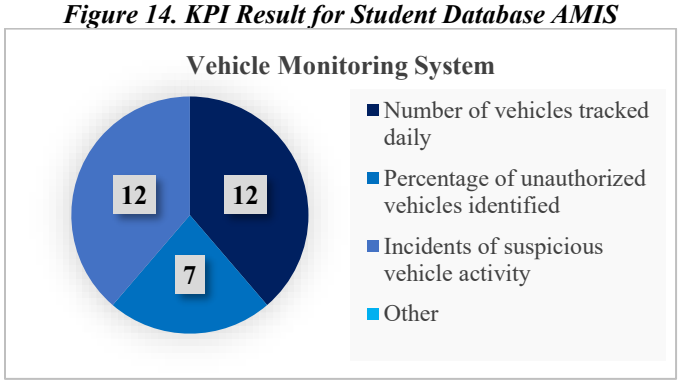


Figure 15. KPI Result for Vehicle Monitoring System

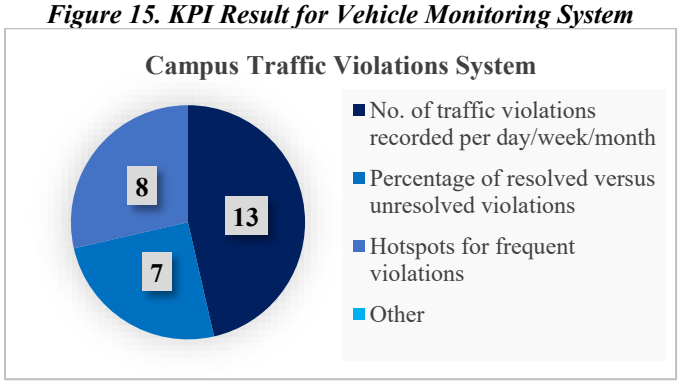


Figure 16. KPI Result for Campus Traffic Violations System

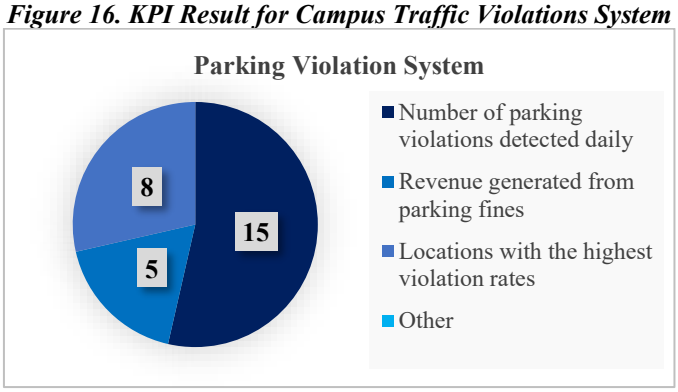


Figure 17. KPI Result for Parking Violations System

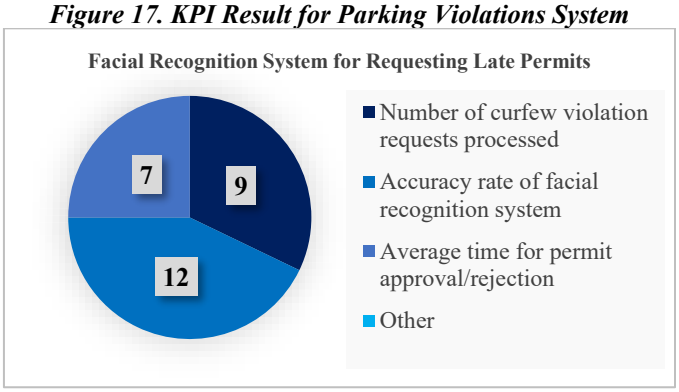


Figure 18. KPI Result for Facial Recognition System for Requesting Late Permits

Reporting Frequency and Statistical Table Outputs

As shown in Figure 19 regarding the report generation, real-time reporting (10 responses) and monthly reporting (10 responses) were preferred, highlighting the need for continuous security monitoring alongside long-term trend analysis. On Figure 20 Security personnel favored visual charts (13 responses) over simple summary tables, suggesting that interactive dashboards with graphical representations would enhance usability and interpretation. The most relevant reporting metrics on Figure 21 included violation trends over time (13 responses) and response time for security incidents (12 responses), ensuring that campus security efforts are quantifiable and actionable.

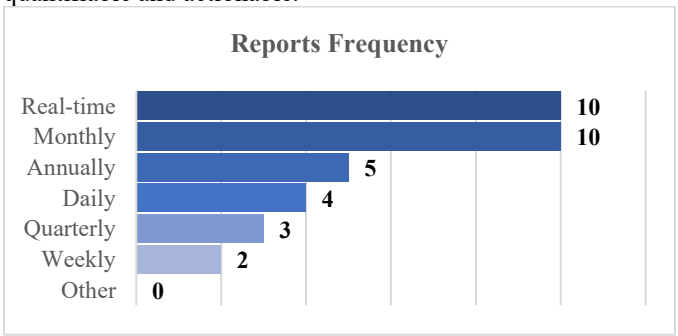


Figure 19. KPI Result for Reporting Frequency

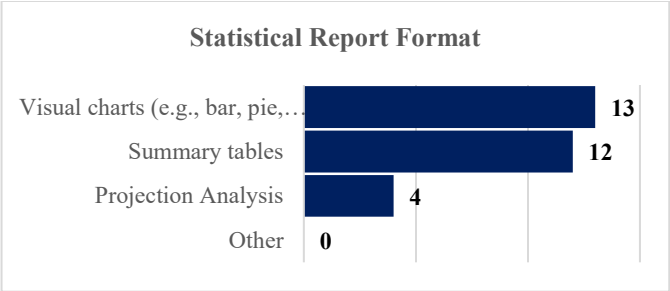


Figure 20. KPI Result for Statistical Report Format

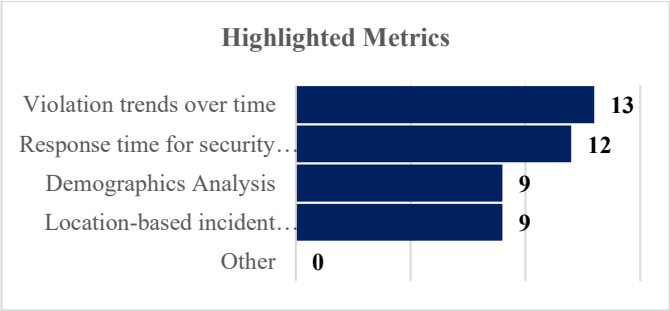


Figure 21. KPI Result for Highlighted Metrics

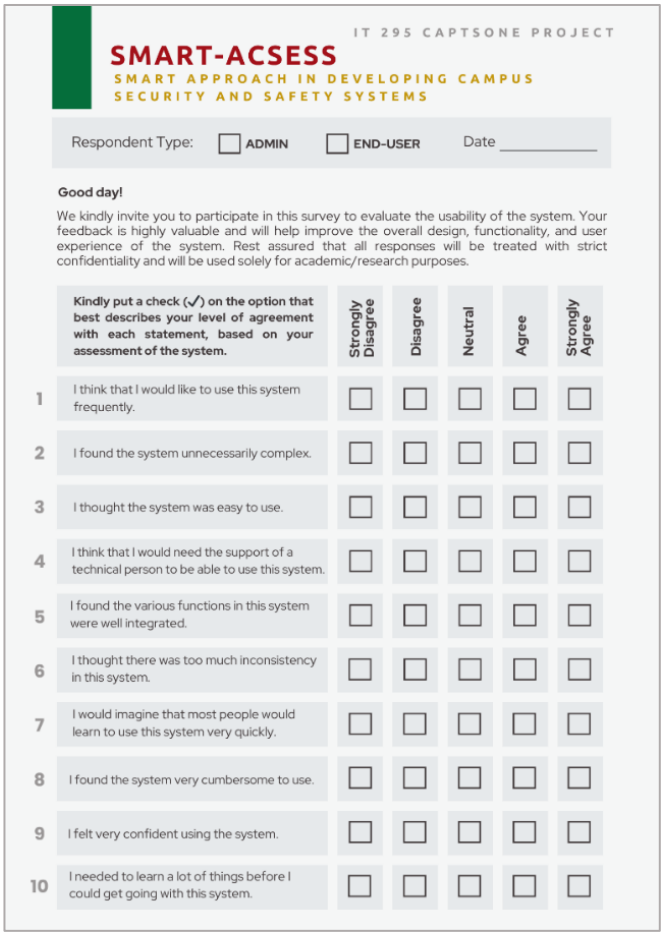
Feedback and Suggestions for Dashboard Optimization

Challenges cited in the feedback section included insufficient security personnel, lack of surveillance technology, absence of automated data analysis, and limited resources. Respondents suggested enhancements such as integrating real-time monitoring technologies, streamlining automated data processing, and improving incident tracking through centralized databases. Additionally, facial recognition, automated violation tracking, and access control systems for university buildings were recommended as valuable additions to the dashboard framework.

The survey findings emphasize the necessity for a data-driven, real-time security dashboard that integrates predictive analytics, interactive visualizations, and automated reporting tools. Security personnel prioritize active vehicle monitoring, biometric-based identification, traffic violation tracking, and enhanced safety measures over AI-driven chatbot functionalities. The feedback underscores the importance of resource allocation, advanced security technologies, and structured statistical analysis in strengthening campus safety initiatives. By leveraging these insights, SMART-ACSESS can refine its centralized security framework to provide an effective and adaptive solution for campus safety management.

iii. SMART-ACSESS Technology Acceptance Model Analysis using the System Usability Scale

SMART-ACSESS adopts the Technology Acceptance Model (TAM) to analyze users perceived usefulness and perceived ease of use of the proposed system, using the System Usability Scale (SUS) in Figure 22, as a standardized instrument to evaluate usability and analyze the following key components: Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Attitude Towards Using, Behavioral Intention (BI), Actual Usage.



IT 295 CAPTSONE PROJECT

SMART-ACSESS

SMART APPROACH IN DEVELOPING CAMPUS SECURITY AND SAFETY SYSTEMS

Respondent Type: ☐ ADMIN ☐ END-USER Date _____

Good day!

We kindly invite you to participate in this survey to evaluate the usability of the system. Your feedback is highly valuable and will help improve the overall design, functionality, and user experience of the system. Rest assured that all responses will be treated with strict confidentiality and will be used solely for academic/research purposes.

	Kindly put a check (✓) on the option that best describes your level of agreement with each statement, based on your assessment of the system.	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Figure 22. SUS Questionnaire for SMART-ACSESS

Based on the results presented in Table 4, three (3) respondents from UPLB-SSO Investigation Unit evaluated the system using the 10-item SUS questionnaire. The computed individual SUS scores are as follows: Respondent 1 obtained a score of 70, Respondent 2 obtained 65, and Respondent 3 obtained 85. The overall average SUS score is 73.33.

According to established SUS interpretation guidelines, an average score of 73.33 falls under the "Good" category, indicating that the system demonstrates above-average usability and is generally acceptable to users. This suggests that the system is perceived as relatively easy to use and functionally adequate in supporting user tasks.

In relation to the Technology Acceptance Model (TAM), the obtained SUS results imply positive user perceptions across key components. The Perceived Ease of Use (PEOU) is reflected in the generally high scores of positively worded items (e.g., ease of use, learnability), indicating that users find the system intuitive and manageable. Similarly, Perceived Usefulness (PU) is supported by users' agreement on the system's functional integration and effectiveness in accomplishing tasks.

Furthermore, favorable usability perceptions contribute to a positive Attitude Towards Using, which in turn strengthens users' Behavioral Intention (BI) to adopt the system. The relatively high scores, particularly from Respondent 3, suggest a strong likelihood of continued usage, thereby supporting Actual Usage of the system in a real-world context.

Overall, the findings indicate that the proposed system achieves a good level of usability, reinforcing its potential for user acceptance and adoption. However, slight variations in individual scores (e.g., Respondent 2) suggest that there is still room for improvement, particularly in enhancing system consistency and reducing perceived complexity.

Q	R1		R2		R3	
	Raw Score	SUS Score	Raw Score	SUS Score	Raw Score	SUS Score
Q1	5	4	5	4	5	4
Q2	5	0	3	2	2	3
Q3	5	4	5	4	4	3
Q4	3	2	5	0	2	3
Q5	5	4	5	4	5	4
Q6	1	4	3	2	1	4
Q7	5	4	5	4	5	4
Q8	3	2	3	2	2	3
Q9	5	4	5	4	4	3
Q10	5	0	5	0	2	3
Total Score (n)		**28**		**26**		**34**
Final Score (n * 2.5)		**70**		**65**		**85**
Average Score	**73.33 (GOOD)**					

Table 4. SUS Result for SMART-ACSESS

V. CONCLUSIONS

This study concludes that SMART-ACSESS provides a viable, scalable, and effective solution to the limitations of traditional campus security systems. By integrating multiple databases into a unified platform supported by real-time APIs, data visualization dashboards, and AI-assisted functionalities, the system enhances operational efficiency, data accessibility, and decision-making capabilities.

Findings from the KPI survey confirm that security personnel prioritize real-time monitoring, vehicle tracking, traffic and parking violation management, and biometric identification systems. These results validate the need for a centralized and data-driven platform that supports proactive and responsive security operations. While chatbot functionalities were ranked lower in priority, they remain valuable as a supplementary tool for improving data accessibility and reporting efficiency.

The usability evaluation, with an average SUS score of 73.33, indicates that the system is generally acceptable and user-friendly, supporting positive perceptions of usefulness and ease of use. This demonstrates strong potential for user adoption and practical implementation within campus security environments. The inclusion of system interface outputs further demonstrates the practical implementation of SMART-ACSESS, validating that the proposed design has been successfully translated into a functional and usable system.

In conclusion, SMART-ACSESS establishes a scalable and future-ready framework for campus security management. Future improvements may include expanded system integration, enhanced predictive analytics, and further optimization of system performance to ensure long-term sustainability and adaptability.

DECLARATION

During the preparation of this work the author(s) used Co-Pilot and ChatGPT in order to improve the readability of the paper including grammar and style checking. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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